



EKSAMENSOPPGÅVE/EKSAMENSOPPGAVE

Emnekode: MAS 149

Emnenamn/Emnenavn: STATIKK OG FASTHETSLÆRE

Utdanning/kull/klasse: Diverse

Dato: 30.09.2025

Eksamensform: Skriftlig

Eksamenstid: 2 timer

Tal på eksamensoppgåver/Antall eksamensoppgaver: 3

Tal på sider (medrekna denne)/Antall sider (inkludert denne): 12

Tal på vedlegg/Antall vedlegg: 5

Tillatne hjelpemiddel/Tillatte hjelpemidler: Enkel kalkulator

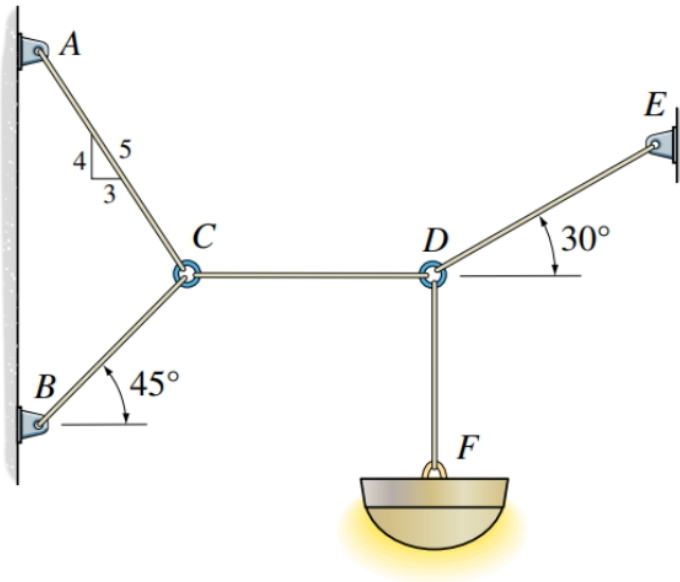
(ikke programmerbar)

Fagansvarleg/Fagansvarlig: Javier de Elio Medina (41259117)

English version

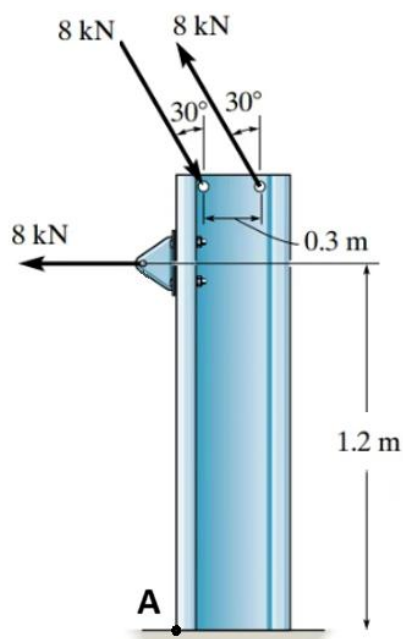
Exercise 1 (30%):

Determine the tension developed in each cord required for equilibrium of the 15-kg lamp



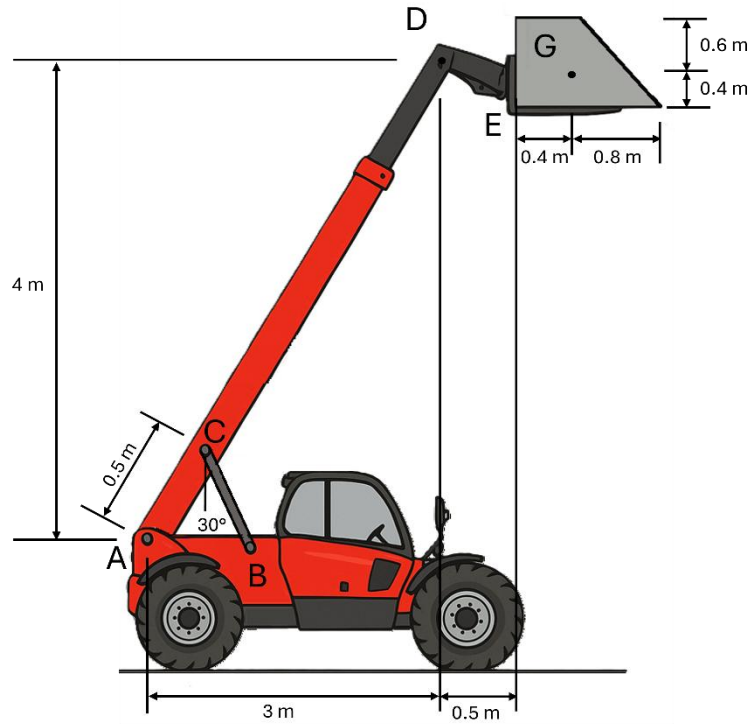
Exercise 2 (30%)

Determine the resultant moment of the forces acting on the beam shown in the figure about point A.



Exercise 3 (40%)

The telescopic handler shown in the following figure is lifting a concrete block of 1500 kg. This load acts at point G. Determine the force in the hydraulic cylinder CB, the support reaction at pin A, and the support reaction at the fixed connection D.

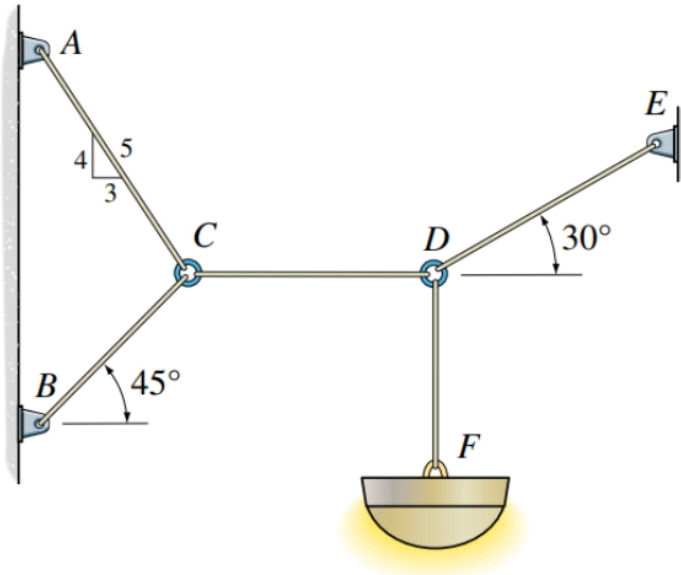


Note: Figure generated with ChatGPT

Bokmål versjon

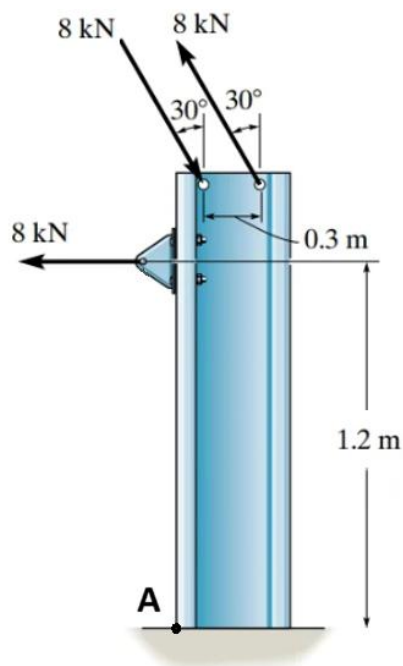
Oppgave 1 (30%):

Finn spenningen som utvikles i hver ledning og som kreves for likevekt i 15 kg-lampen.



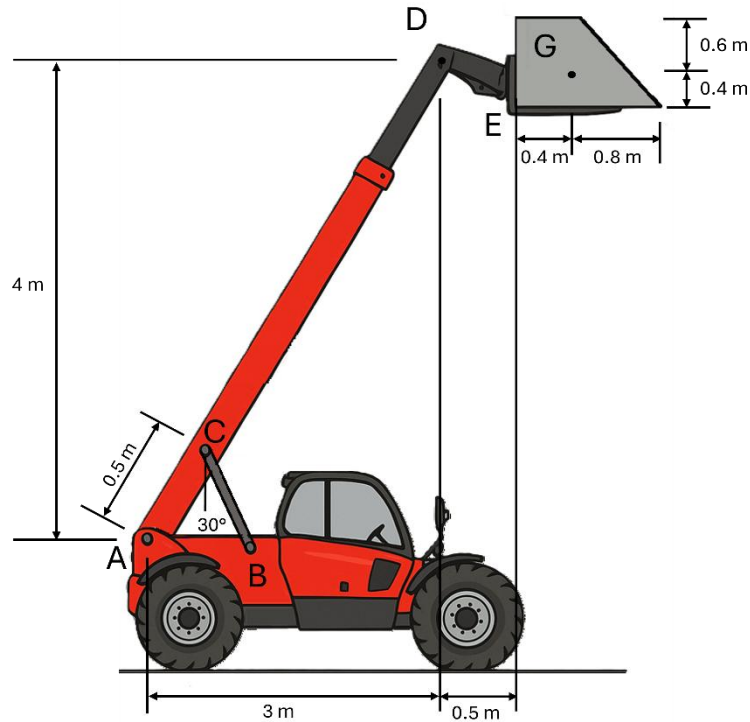
Oppgave 2 (30%)

Finn det resulterende kraftmomentet til kreftene som virker på bjelken vist i figuren om punkt A .



Oppgave 3 (40%)

Teleskoplasteren vist i figuren nedenfor løfter en homogen betongblokk på 1500 kg. Denne lasten virker i punkt G. Finn kraften i den hydrauliske sylindren CB, reaksjonskreftene i boltelager A, og reaksjonskreftene i den faste innspenningen D.

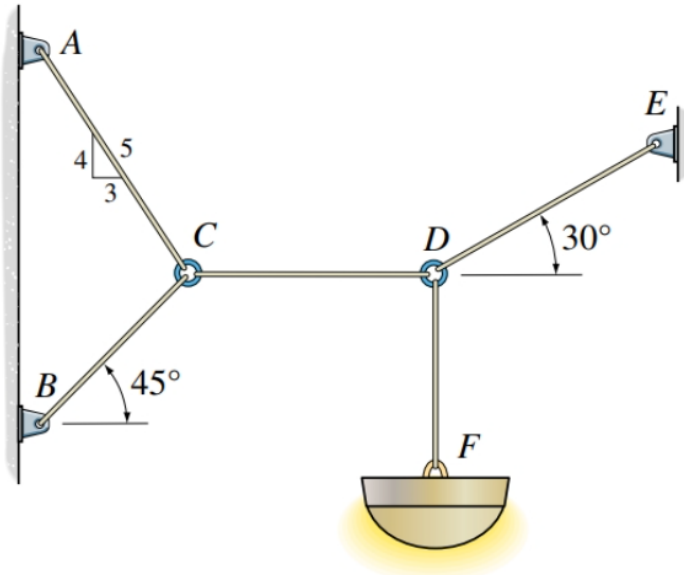


Note: Figuren er generert av ChatGPT

Nynorsk versjon

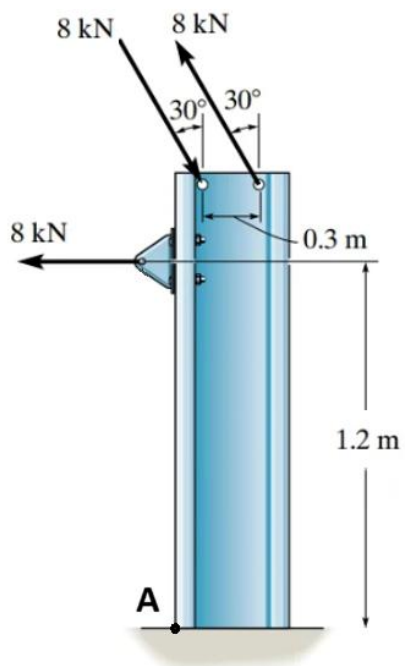
Oppgåve 1 (30%):

Finn spenninga som blir utvikla i kvar leidning og som krevst for likevekt i 15 kg lampen



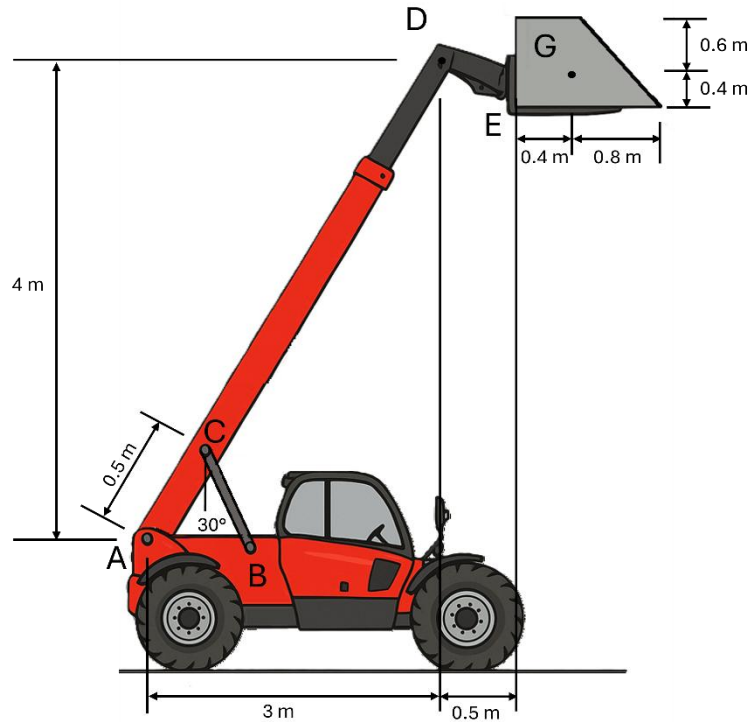
Oppgåve 2 (30%)

Finn resulterande kraftmomentet til kreftene som verkar på bjelken vist i figuren om punkt A.



Oppgave 3 (40%)

Teleskoplasteren vist i figuren nedanfor løfte ei homogen betongblokk på 1500 kg. Denne lasta verkar i punkt G. Finn krafta i den hydrauliske sylindern CB, reaksjonskreftene i boltelager A, og reaksjonskreftene i den faste innspenninga D.



Note: Figuren er generert av ChatGPT

Eksamen i MAS149 Statikk og fasthetslære, vedlegg

Friksjon:

$$F_{max} = \mu_s \cdot P$$

Varigions prinsipp

$$M_O = \Sigma F_x \cdot y - \Sigma F_y \cdot x = F_R \cdot a$$

Resultantkraft

$$F_R = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$$

Aksial spenning:

$$\sigma_{avg.} = \frac{P}{A}$$

Elastisk forlengelse:

$$\delta_F = \frac{P \cdot L}{A \cdot E}$$

$$\delta_T = \alpha \cdot \Delta T \cdot L$$

Torsjon:

$$\tau = \frac{T \cdot c}{J}$$

sirkulære tverrsnitt:

$$\text{massivt: } J = \frac{\pi}{2} c^4$$

$$\text{rør: } J = \frac{\pi}{2} (c_y^4 - c_i^4)$$

Vridningsvinkel, ϕ

$$\phi = \int_0^L \frac{T(x)dx}{J(x)G}$$

$$\phi = \Sigma \frac{TL}{JG}$$

Bøying:

$$\sigma_b = \frac{M \cdot c}{I}$$

Sammenheng mellom w, V og M:

$$\frac{dV}{dx} = -w \quad \frac{dM}{dx} = V$$

Symbol	Forklaring	Benevning
F_{max}	maksimal friksjonskraft	[N]
F_R	resultantkraft	[N]
y, x, a	momentarmen til henholdsvis F_y , F_x og F_R	[m]
μ_s	statisk friksjonskoeffisient	
P	normalkraft	[N]
σ_{avg}	gjennomsnittlig normalspenning	[Pa]
σ_b	bøyespenning	[Pa]
σ_s	totale strekkspenninger (bøye og normalspenninger)	[Pa]
σ_t	totale trykkspenninger (bøye og normalspenninger)	[Pa]
d	diameter	[m]
A	tverrsnittsareal	[m ²]
E	elastisitetsmodul	[Pa]
δ_F	defleksjon p.g.a. av ytre aksiallast	[m]
δ_T	defleksjon p.g.a. temperaturforandring	[m]
ΔT	temperaturforandring	[°C ⁻¹]
τ_{max}	maksimal torsjon og skjærspenning	[Pa]
$\tau_{avg.}$	gjennomsnittlig skjærspenning	[Pa]
T	torsjonsmoment	[Nm]
c	avstand fra t.p. til ytterkant i tverrsnittet	[m]
c_i og c_y	indre og ytre radius	[m]
J	polart arealmoment	[m ⁴]
M	bøymoment	[Nm]
w	last pr. meter (kan også være vist som q)	[N/m]
I	annet arealmoment	[m ⁴]
I'	annet arealmoment med Steiners formel	[m ⁴]
d_y og d_x	avstand til tyngdepunkt	[m]
V	skjærkraft	[N]
Q	statisk moment	[m ³]
A'	arealet av flate elementet til tverrsnittet	[m ²]
$\bar{y}'$	avst. fra t.p. i tverrsnitt til t.p i A'	[m]
t	tykkelse	[m]
P_{cr}	maks. last søylen tåler før den knekker	[N]
σ_{cr}	maks. spenning søylen tåler før den knekker	[Pa]
L	lengden på søylen	[m]
K	effektiv lengdefaktor	
r	tregghetsradius (radius of gyration)	[m]
FS	sikkerhetsfaktor (Factor of Safety)	
$S_{req'd}$	nødvendig motstandsmoment (section modulus)	[m ³]
	Pa = N/m ²	
	MPa = N/mm ²	
	GPa = 10 ³ Mpa	
	1 m = 10 ³ mm	
	1 m ² = 10 ⁶ mm ²	
	1 m ³ = 10 ⁹ mm ³	

Dimensjonering:

$$S_{req'd} = \frac{M_{max.}}{\sigma_{allow}} = \frac{I}{c}$$

Skjær:

$$\tau_{avg.} = \frac{V}{A} \quad \tau_{max.} = \frac{V \cdot Q}{I \cdot t} \quad Q = \sum \bar{y} \cdot A'$$

Parallell-akse teoremet (Steiner's formel)

$$I'_x = I_x + A \cdot d_y^2$$

$$I'_y = I_y + A \cdot d_x^2$$

Tyngdepunkt

$$\bar{x} = \frac{x_1 \cdot A_1 + x_2 \cdot A_2 + \dots + x_n \cdot A_n}{A_1 + A_2 + \dots + A_n}$$

$$\bar{y} = \frac{y_1 \cdot A_1 + y_2 \cdot A_2 + \dots + y_n \cdot A_n}{A_1 + A_2 + \dots + A_n}$$

Kombinert spenning

$$\text{Cylinder} \quad \sigma_1 = \frac{pr}{t} \quad \sigma_2 = \frac{pr}{2t} \quad \text{Sphere} \quad \sigma_1 = \sigma_2 = \frac{pr}{2t}$$

Normalspenninger

$$\sigma_s = \sigma_b + \sigma_n \quad \sigma_t = \sigma_b - \sigma_n$$

Spenningstransformasjonsligninger

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta \quad \tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

$$\sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y$$

Hovedspenninger

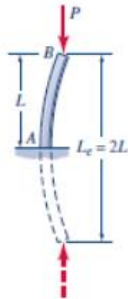
$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad \tan 2\theta_p = \frac{\tau_{xy}}{(\sigma_x - \sigma_y)/2}$$

Maks. plan skjærspenning

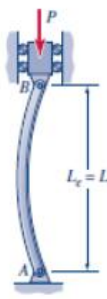
$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad \tan 2\theta_s = -\frac{(\sigma_x - \sigma_y)/2}{\tau_{xy}}$$

Knekking:

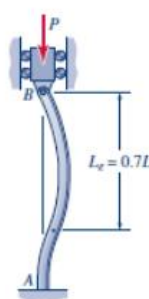
$$P_{cr} = \frac{\pi^2 \cdot E \cdot I}{(K \cdot L)^2} \quad \sigma_{cr} = \frac{\pi^2 \cdot E}{\left(\frac{K \cdot L}{r}\right)^2} \quad r = \sqrt{\frac{I}{A}} \quad FS = \frac{P_{cr}}{P}$$



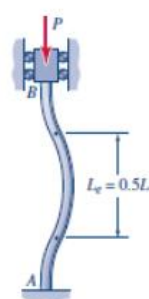
a) $K = 2$



b) $K = 1$



c) $K = 0,7$



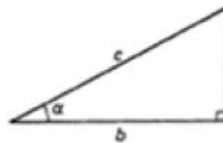
d) $K = 0,5$

- a) Stav med den ene enden innspent og den andre enden fri.
- b) Stav med begge ender leddet.
- c) Stav med den ene enden innspent og den andre enden leddet.
- d) Stav med begge endene innspent.

TRIGONOMETRI

Pytagoreiske
læresetning

$$c^2 = a^2 + b^2$$



c hypotenus
 a |
 b kateter

Trigonometriske
funksjoner

$$\sin \alpha = \frac{a}{c} \quad \cos \alpha = \frac{b}{c} \quad \tan \alpha = \frac{a}{b}$$

Sinussetningen

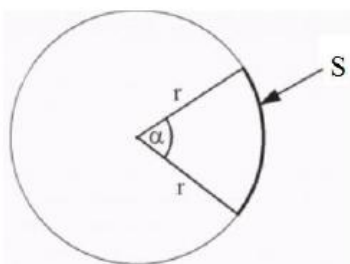
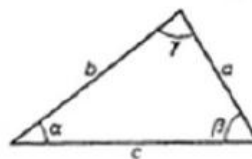
$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$$

Cosinussetningen

$$a^2 = b^2 + c^2 - 2bc \cos \alpha$$

Tangenssetningen

$$\frac{a-b}{a+b} = \frac{\tan \frac{\alpha-\beta}{2}}{\tan \frac{\alpha+\beta}{2}}$$



$$S = \alpha \cdot r \quad \alpha [\text{rad}]$$

Fundamental Equations of Mechanics of Materials

Axial Load

Normal Stress

$$\sigma = \frac{N}{A}$$

Displacement

$$\delta = \int_0^L \frac{N(x)dx}{A(x)E}$$

$$\delta = \Sigma \frac{NL}{AE}$$

$$\delta_T = \alpha \Delta TL$$

Torsion

Shear stress in circular shaft

$$\tau = \frac{T\rho}{J}$$

where

$$J = \frac{\pi}{2} c^4 \quad \text{solid cross section}$$

$$J = \frac{\pi}{2} (c_o^4 - c_i^4) \quad \text{tubular cross section}$$

Power

$$P = T\omega = 2\pi fT$$

Angle of twist

$$\phi = \int_0^L \frac{T(x)dx}{J(x)G}$$

$$\phi = \Sigma \frac{TL}{JG}$$

Average shear stress in a thin-walled tube

$$\tau_{\text{avg}} = \frac{T}{2tA_m}$$

Shear Flow

$$q = \tau_{\text{avg}} t = \frac{T}{2A_m}$$

Bending

Normal stress

$$\sigma = -\frac{My}{I}$$

Unsymmetric bending

$$\sigma = -\frac{M_z y}{I_z} + \frac{M_y z}{I_y}, \quad \tan \alpha = \frac{I_z}{I_y} \tan \theta$$

Shear

Average direct shear stress

$$\tau_{\text{avg}} = \frac{V}{A}$$

Transverse shear stress

$$\tau = \frac{VQ}{It}$$

Shear flow

$$q = \tau t = \frac{VQ}{I}$$

Stress in Thin-Walled Pressure Vessel

Cylinder

$$\sigma_1 = \frac{pr}{t} \quad \sigma_2 = \frac{pr}{2t}$$

Sphere

$$\sigma_1 = \sigma_2 = \frac{pr}{2t}$$

Stress Transformation Equations

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

Principal Stress

$$\tan 2\theta_p = \frac{\tau_{xy}}{(\sigma_x - \sigma_y)/2}$$

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

Maximum in-plane shear stress

$$\tan 2\theta_s = -\frac{(\sigma_x - \sigma_y)/2}{\tau_{xy}}$$

$$\tau_{\text{max}} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2}$$

Absolute maximum shear stress

$$\tau_{\text{abs max}} = \frac{\sigma_{\text{max}}}{2} \text{ for } \sigma_{\text{max}}, \sigma_{\text{min}} \text{ same sign}$$

$$\tau_{\text{abs max}} = \frac{\sigma_{\text{max}} - \sigma_{\text{min}}}{2} \text{ for } \sigma_{\text{max}}, \sigma_{\text{min}} \text{ opposite signs}$$

Geometric Properties of Area Elements

Material Property Relations

Poisson's ratio

$$\nu = -\frac{\epsilon_{\text{lat}}}{\epsilon_{\text{long}}}$$

Generalized Hooke's Law

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)]$$

$$\epsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)]$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)]$$

$$\gamma_{xy} = \frac{1}{G} \tau_{xy}, \gamma_{yz} = \frac{1}{G} \tau_{yz}, \gamma_{zx} = \frac{1}{G} \tau_{zx}$$

where

$$G = \frac{E}{2(1 + \nu)}$$

Relations Between w , V , M

$$\frac{dV}{dx} = w(x), \quad \frac{dM}{dx} = V$$

Elastic Curve

$$\frac{1}{\rho} = \frac{M}{EI}$$

$$EI \frac{d^4 v}{dx^4} = w(x)$$

$$EI \frac{d^3 v}{dx^3} = V(x)$$

$$EI \frac{d^2 v}{dx^2} = M(x)$$

Buckling

Critical axial load

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

Critical stress

$$\sigma_{cr} = \frac{\pi^2 E}{(KL/r)^2}, r = \sqrt{I/A}$$

Secant formula

$$\sigma_{max} = \frac{P}{A} \left[1 + \frac{ec}{r^2} \sec \left(\frac{L}{2r} \sqrt{\frac{P}{EA}} \right) \right]$$

Energy Methods

Conservation of energy

$$U_e = U_i$$

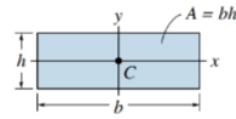
Strain energy

$$U_i = \frac{N^2 L}{2AE} \quad \text{constant axial load}$$

$$U_i = \int_0^L \frac{M^2 dx}{2EI} \quad \text{bending moment}$$

$$U_i = \int_0^L \frac{f_s V^2 dx}{2GA} \quad \text{transverse shear}$$

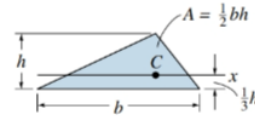
$$U_i = \int_0^L \frac{T^2 dx}{2GJ} \quad \text{torsional moment}$$



Rectangular area

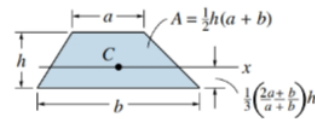
$$I_x = \frac{1}{12} bh^3$$

$$I_y = \frac{1}{12} hb^3$$

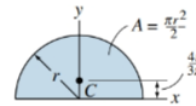


Triangular area

$$I_x = \frac{1}{36} bh^3$$



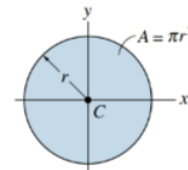
Trapezoidal area



Semicircular area

$$I_x = \frac{1}{8} \pi r^4$$

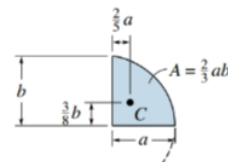
$$I_y = \frac{1}{8} \pi r^4$$



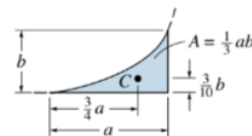
Circular area

$$I_x = \frac{1}{4} \pi r^4$$

$$I_y = \frac{1}{4} \pi r^4$$



Semiparabolic area



Exparabolic area